

Herdi Saleh

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SKILLS

- **Languages**
 - Swedish (Native), English (Fluent), Kurdish (Fluent)
- **Programming Languages**
 - Haskell, Java, C, C++, Python 3, SQL
- **Technologies**
 - git, Jira, Gerrit, Jenkins, GitHub, GNU/Linux, LaTeX, CMake, Visual Studio Code, tmux, PostgreSQL
- **Other**
 - Swedish Citizen, Driver's License (B-körkort)

WORK EXPERIENCE

Compiler Thesis Student

January 2026 – June 2026

Oracle

Stockholm, Sweden

- Developed a semantic framework to formalize the memory operations of the Sea of Nodes intermediate representation in the HotSpot JVM C2 compiler.
- Used Haskell, interfacing with Z3, to build a translation validation tool to verify C2 compiler optimizations using the developed semantic framework.
- Developed an accompanying fuzzer to generate Java programs as input for the tool.

Compiler Team Intern

June 2024 – August 2025

Ericsson

Stockholm, Sweden

- Developed and maintained a fuzz testing tool in Haskell for an in-house compiler built on Clang/LLVM.
- Applied techniques from state-of-the-art research papers (CSmith, YARPGen v.1/v.2) to generate C23 compliant code with extensions specific to Ericsson DSP.
- Overall, fuzzing efforts identified **16+** compiler bugs, with individual contributions focused on extending the template system to fuzz more compiler optimizations, extending the internal constraint-based type system, etc.
- Worked in a 6-person team using Agile (Scrum) for sprint-based feature delivery.

Front Desk Receptionist

July 2023 – August 2023

Vasastans vårdcentral

Stockholm, Sweden

- Managed patient interactions, greeting and registering up to 30 patients daily, ensuring a welcoming and professional atmosphere.

Volunteering Tutor

March 2022 – May 2022

Stockholms stad

Stockholm, Sweden

- Tutored in all subjects for students studying in middle school, high school, and Komvux.
- Improved communication skills and adaptability to meet students' needs.

EDUCATION

KTH Royal Institute of Technology

August 2021 – July 2026

Master of Science in CSE, sv. Civilingenjör i Datateknik

Stockholm, Sweden

- Software technology track, with specialization in programming languages.

SCHOLARSHIPS AND AWARDS

Scholarship for good study results

June 2025

Stiftelsen Henrik Göranssons Sandviken, KTH